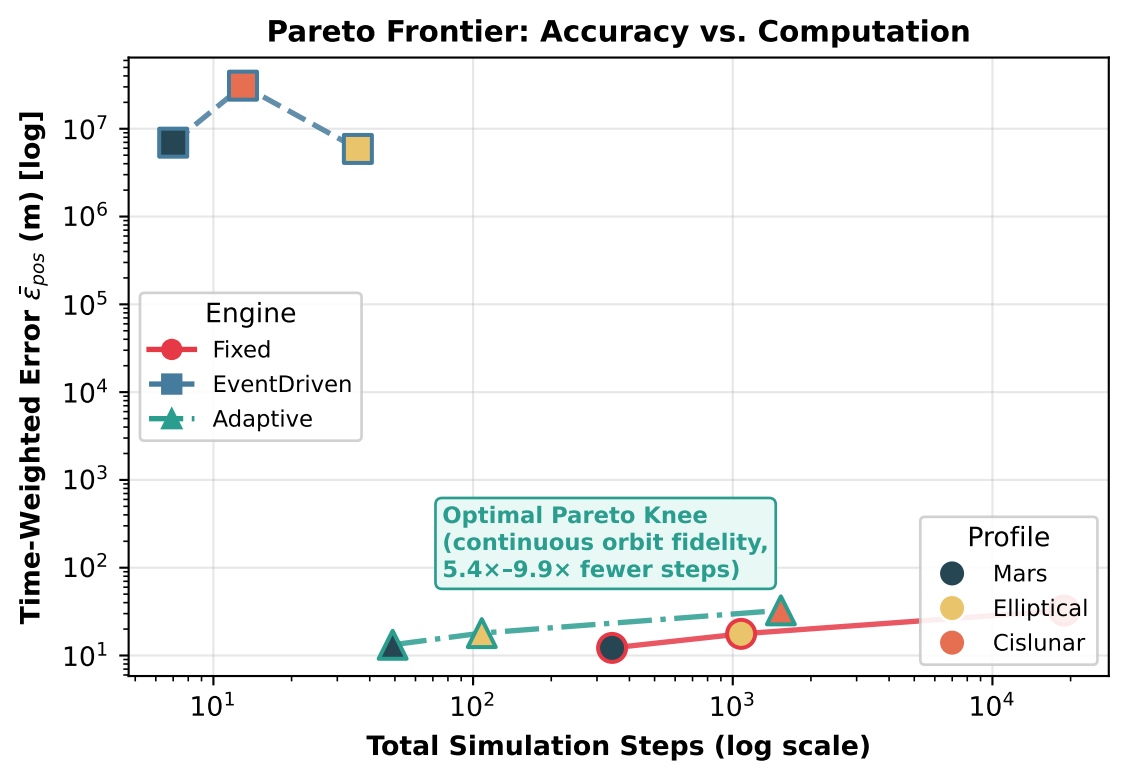
Pareto Frontier: Accuracy vs. Computation
Time-Weighted Error $\bar{\varepsilon}_{pos}$ (m) [log]
Total Simulation Steps (log scale)
10^7
10^6
10^5
10^4
10^3
10^2
10^1
10^1
10^2
10^3
10^4
Engine
Fixed
EventDriven
Adaptive
Optimal Pareto Knee
(continuous orbit fidelity,
5.4×–9.9× fewer steps)
Profile
Mars
Elliptical
Cislunar